#WebTransport

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-> New technology.
-> websocket vs) webTransport (Replacement ??)

* Intro
-> webTransport is a new web communication protocol built on HTTP/3 (QUIC)

• Benefits
-> Prevent (Head of line blocking issues)
->TCP based had to resched the packet if lost, resulting in multiple request chek.

-> Build on top of HTTP/3 quic
-> Maintains bidirectional messaging like web sockets.
-> Supports both TCP/UDP

-> Browsers support is increasing

* WebSocket
-> Bidirectional communication (improvement over polling and server sent events)

-> Lower overhead (no HTTP headers after hand shake)
-> Real-time (TCP)
-> Persistance communication

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✓✓ Websockets Limitations
-> Head of Line Problems
-> Singl stream per connection
• If youre sending different types of data, large message can clog the pipe for other data types.
-> Reliability constraints
• Always ensures reliable, ordered delivery Ineffecient for live video or game state.

Request with TCP + TLS (HTTP/2)

-> Three way Handshake
-> TLS (creates) a separate Tunnel Secure
-> Upgrade to websocket
-> Websocket conn, established.
• Uses single Bidirectional message channel

Request with QUIC
-> Three way QUIC Handshake (includes TLS)
-> Upgrade to webTransport
-> webTransport connection established.

• Less Round Trips
• UDP

• Uses same UDP conn for both types of connection, Streams and datagrams.

we can have both reliable and non-reliable datachannel within single UDP connection.

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WebTransports

-> Secured with (TLS)
-> Faster handshake
-> Supports both Streams and Datagrams within a single connection.
-> Multiplexed Streams (uses different channel for different data types)

* Connection Migration
• Users / device might move from one type of connection to another. For example, wifi to LTE (Telecom)

↳ If the connection was established with websocket, then on network changed, the connection has to reestablished from the begins (Different IP address, as network changes)

↳ But, webTransport (QUIC) uses webTranspo Connection ID instead of ip addresses, so when we switch network and get a new ip address the webTransport automatica reconnects to the server using the same conne ID.

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* Real world Examples

- High Frequency Trading Platforms
  - → Market data via UDP (latest price matters)
  - → Orders confirmations (TCP reliable stream)
  - → Banking (reliable stream)
  - → WebTransport wins, over WebSocket

- Cloud Gaming
  - → Video transfer [UDP]
  - → Player input
  - → WebTransport wins

- Real-Time collaborative Design Tools
  - → cursor movements (UDP, for real time)
  - → Document changes (TCP, every change should be preserved)
  - → Voice chat (UDP)
  - → WebTransport wins:

* When to use which

- WebSocket
  - → simple real-time messaging (chat, noti,)
  - → Universal browser support required
  - → Single data type
  - → Reliable delivery essential

- WebRTC
  - → Peer to peer connection
  - → Real-time audio, video
  - → No server bandwidth costs (P2P file sharing)

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- WebTransport
-> High performance client - server
-> Tuned reliability requirements
-> Multiple independent data streams
-> Ultra low latency applications
->

How Implementation works in real world.

In real world application, we should not rely on single protocol.

So, the basic principle is:

-> Use webTransport
  - Safari no support
  - corporate network blocks QUIC (Firewall, Packet inspect)

(> If these, then

-> Fall Back to web sockets
  - Old browers don't support
  - Corporate network can block WS (Firewall, Packet inspect)

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  .
(> If these, then

-> Fall back to server sent Events.

So, in real world applications.

Graceful Fallback strategy is used.